Solutions to Dummit and Foote

Labix

July 25, 2026

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1 Introduction to Groups

1.1 Basic Axioms and Examples

Exercise 1.1.1

(a) Not associative. $(2-3)-4=-5$ and $2-(3-4)=3$ are different.
(b) Associative. We have

$$(a \star b) \star c = (a+b+ab) \star c = a+b+c+ab+ac+bc+abc = a \star (b+c+bc) = a \star (b \star c)$$

since addition and multiplication in $\mathbb{R}$ is associative.
(c) Not associative. $(1 \star 2) \star 3 = \frac{18}{25}$ and $1 \star (2 \star 3) = \frac{2}{5}$ are different.
(d) Associative. We have $((a,b) \star (c,d)) \star (e,f) = (ad+bc, bd) \star (e,f) = (adf+bcf+bde, bdf)$ and $(a,b) \star ((c,d) \star (e,f)) = (a,b) \star (cf+de, df) = (adf+bcf+bde, bdf)$ since addition and multiplication in $\mathbb{Z}$ is associative.
(e) Not associative. $1/(2/3) = 3/2$ and $(1/2)/3 = 1/6$ are different.

Exercise 1.1.2

(a) Not commutative. $-1 = 1-2 \neq 2-1 = 1$.
(b) Commutative. We have $a \star b = a+b+ab = b+a+ba = b \star a$ since addition and multiplication in $\mathbb{R}$ is commutative.
(c) Commutative. We have $\frac{a+b}{5} = \frac{b+a}{5}$ since addition in $\mathbb{Q}$ is commutative.
(d) Commutative. We have

$$(a,b) \star (c,d) = (ad+bc, bd) = (cb+da, db) = (c,d) \star (a,b)$$

since addition and multiplication in $\mathbb{Z}$ is commutative.
(e) Not commutative. $1/2 \neq 2/1 = 2$.

Exercise 1.1.3

Let $0 \leq i,j,k \leq n-1$. Then there exists $0 \leq t \leq n-1$ such that $i+j = an+t$ for some $a \in \mathbb{Z}$. Similarly, there exists $0 \leq s \leq n-1$ such that $t+k = bn+s$ for some $b \in \mathbb{Z}$. Then by definition we have

$$(\bar{i}+\bar{j})+\bar{k} = \bar{t}+\bar{k} = \bar{s}$$

Also we have $i+j+k = an+bn+s$. On the other hand, there exists $0 \leq x,y \leq n-1$ such that $j+k = cn+x$ and $i+x = dn+y$ for some $c,d \in \mathbb{Z}$. Then by definition we have

$$\bar{i}+(\bar{j}+\bar{k}) = \bar{i}+\bar{x} = \bar{y}$$

and $i+j+k = cn+dn+y$. Hence $an+bn+s = i+j+k = cn+dn+y$. This implies that $(a+b-c-d)n+s = y$ so that $\bar{s} = \bar{y}$. Thus we are done.

Exercise 1.1.4

Let $0 \leq i,j,k \leq n-1$. Then there exists $0 \leq t \leq n-1$ such that $ij = an+t$ for some $a \in \mathbb{Z}$. Similarly, there exists $0 \leq s \leq n-1$ such that $tk = bn+s$ for some $b \in \mathbb{Z}$. Then by definition we have

$$(\bar{i}\bar{j})\bar{k} = \bar{t}\bar{k} = \bar{s}$$

Also we have $ijk = ank+bn+s$. On the other hand, there exists $0 \leq x,y \leq n-1$ such that $jk = cn+x$ and $ix = dn+y$ for some $c,d \in \mathbb{Z}$. Then by definition we have

$$\bar{i}(\bar{j}\bar{k}) = \bar{i}\bar{x} = \bar{y}$$

and $ijk = cni+dn+y$. Hence $ank+bn+s = i+j+k = cni+dn+y$. This implies that $(ak+b-ci-d)n+s = y$ so that $\bar{s} = \bar{y}$. Thus we are done.

Exercise 1.1.5
I claim that $\overline{0}$ has no multiplicative inverse. Suppose that $\overline{0}\overline{x} = \overline{1}$ for some $0 \leq x \leq n-1$. By definition of multiplication, there exists $k \in \mathbb{Z}$ such that $1 - 0x = kn$. But this implies that $k = 1/n$ which is a contradiction to $k \in \mathbb{Z}$ since $n > 1$.

1.2

1.3

1.4 Matrix Groups

Exercise 1.4.1
Clearly we have $|M_2(\mathbb{F}_2)| = 8$. But $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ if and only if $ad - bc = 1$. Then we have $ad = 0$ and $bc = 1$, or $ad = 1$ and $bc = 0$. The first case gives $b = c = 1$ and $a = 0$ or $d = 0$ which is three possibilities. By symmetry the latter also gives three possibilities. Hence $|\mathrm{GL}_2(\mathbb{F}_2)| = 6$.

Exercise 1.4.2
By the above exercise, we have the following six elements of $\mathrm{GL}_2(\mathbb{F}_2)$.

- The element $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ satisfies $A^2 = I$. Hence $|A| = 2$.
- The element $B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ satisfies $B^2 = C$ and $B^3 = I$. Hence $|B| = 3$.
- The element $C = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ satisfies $C^2 = B$ and $C^3 = I$. Hence $|C| = 3$.
- The element $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is just the identity.
- The element $E = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ satisfies $E^2 = I$. Hence $|E| = 2$.
- The element $F = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ satisfies $F^2 = I$. Hence $|F| = 2$.

Exercise 1.4.3
A simple calculation shows that $AE = C$ but $EA = B$.

Exercise 1.4.4
Suppose that $n = ab$ for some $a, b \in \mathbb{N}$ such that $a, b \neq 1$. I claim that $\overline{a}$ has no inverse element. Suppose for a contradiction that $\overline{x}\overline{a} = \overline{1}$ for some $0 \leq x \leq n-1$. Multiplying by $\overline{b}$ on both sides give

$$\overline{0} = \overline{x}\overline{0} = \overline{x}\overline{a}\overline{b} = \overline{b}$$

Hence b is divisible by n. But $b \leq n$ implies that $b = n$ and $a = 1$. This is a contradiction.

Exercise 1.4.5
If F is finite, then $|M_n(F)| = |F|^{n^2}$ and $\mathrm{GL}_n(F) \subseteq M_n(F)$ implies that it is finite. If F is infinite, then $\left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \;\middle|\; x \in F \right\}$ is an infinite set of elements contained in $\mathrm{GL}_n(F)$.

Exercise 1.4.6
We know that $\mathrm{GL}_n(F) \subseteq M_n(F)$ which implies that $|\mathrm{GL}_n(F)| \leq q^{n^2}$. But the 0 matrix is non-invertible. Hence $|\mathrm{GL}_n(F)| \leq q^{n^2} - 1 < q^{n^2}$.

Exercise 1.4.7
There are three cases to consider.

- Case 1: The first row is 0. There are p^2 choices for the two elements in the last row.

- Case 2: The last row is 0. There are p^2 choices for the two elements in the first row.
- Case 3: The remaining case. Fix the top row. Then there are (p^2-1) choices so that not both of them are 0. There are $(p-1)$ choices for the scaling factor so that not both of the elements in the last row is 0. Since the scaling factor is non-zero, the first row is automatically a non-zero multiple of the second row.

Considering that we double counted the 0 matrix, the number of matrices whose one row is a multiple of the other is given by $p^2+p^2+(p-1)(p^2-1)-1 = p^3+p^2-p$. Hence we have

$$|\mathrm{GL}_n(\mathbb{F}_p)| = p^4-p^3-p^2+p$$

Exercise 1.4.8

Using block matrix notation and notation from ex1.4.2, notice that

$$\begin{pmatrix} A & 0 \\ 0 & I_{n-2} \end{pmatrix}\begin{pmatrix} E & 0 \\ 0 & I_{n-2} \end{pmatrix} = \begin{pmatrix} C & 0 \\ 0 & I_{n-2} \end{pmatrix} \neq \begin{pmatrix} B & 0 \\ 0 & I_{n-2} \end{pmatrix} = \begin{pmatrix} E & 0 \\ 0 & I_{n-2} \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & I_{n-2} \end{pmatrix}$$

Exercise 1.4.9

1.5

1.6 Homomorphisms and Isomorphisms

Exercise 1.6.1

(a) We induct on n. When $n=1$ this is trivial. Now suppose that $\varphi(x^k)=\varphi(x)^k$ for some $k\in\mathbb{N}^+$. Then we have

$$\varphi(x^k) = \varphi(x^k\cdot x) = \varphi(x^k)\varphi(x) = \varphi(x)^k\varphi(x) = \varphi(x)^{k+1}$$

so we are done.

(b) We claim that $\varphi(x^{-1})$ is the inverse of $\varphi(x)$. Indeed, we have

$$\varphi(x)\varphi(x^{-1}) = \varphi(x\cdot x^{-1}) = \varphi(1_G) = 1_H$$

where $\varphi(1_G)=\varphi(1_H)$ because $\varphi(1_G)=\varphi(1_G\cdot 1_G)=\varphi(1_G)\varphi(1_G)$ so applying $\varphi(1_G)^{-1}$ to both sides give $\varphi(1_G)=1_H$. Now we have

$$\varphi(x)^{-n} = (\varphi(x)^{-1})^n = \varphi(x^{-1})^n = \varphi((x^{-1})^n) = \varphi(x^{-n})$$

for any $n\in\mathbb{N}$. So we are done.

Exercise 1.6.2

Suppose that $|x|$ is finite. Then we have

$$1_H = \varphi(1_G) = \varphi(x^{|x|}) = \varphi(x)^{|x|}$$

Hence we have $|\varphi(x)|\le|x|$. Similarly, since $|\varphi(x)|$ is finite, we can apply the same argument to φ^{-1} to show that $|x|\le|\varphi(x)|$. Thus $|x|=|\varphi(x)|$. Now suppose that $|x|$ is not finite. Suppose for a contradiction that $\varphi(x)^n=1_H$ for some $n\in\mathbb{N}^+$. Then we have

$$\varphi(x^n) = \varphi(x)^n = 1_H = \varphi(1_G)$$

Since φ is an isomorphism, it is injective and $x^n=1_G$, contradicting the fact that $|x|$ is not finite. Thus $|\varphi(x)|$ is not finite.

The above shows that φ and φ^{-1} establishes a bijection between elements of G and H with fixed order.

The result is no longer true. Consider the homomorphism $\varphi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ given by $n \mapsto [n]$. The element 1 in $\mathbb{Z}$ has infinite order but $[1] \in \mathbb{Z}/2\mathbb{Z}$ has order 2.

Exercise 1.6.3

Suppose that G is abelian. Let $h_1, h_2 \in H$. Since φ is an isomorphism, there exists $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Then we have

$$h_1h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) = \varphi(g_2g_1) = \varphi(g_2)\varphi(g_1) = h_2h_1$$

Thus H is abelian. The result can be applied for φ^{-1} to deduce that if H is abelian, then G is abelian.

The above argument only relies on the assumption that φ is surjective.

Exercise 1.6.4

By Q2, The number of elements of a fixed order is the same number between any isomorphic groups. There are no element of order 3 in $\mathbb{R}^\times$ but there are two in $\mathbb{C}^\times$ given by $e^{2\pi i/3}$ and $e^{4\pi/3}$.

Exercise 1.6.5

Suppose that $\varphi : \mathbb{Q} \to \mathbb{Z}$ is an isomorphism. Then for any $n \in \mathbb{N}^+$, we have $n \cdot \varphi(1/n) = \sum_{i=1}^{n} \varphi(1/n) = \varphi\left(\sum_{i=1}^{n} 1/n\right) = \varphi(1)$. Thus n divides $\varphi(1)$ for all n, a contradiction.

Exercise 1.6.6

Notice that D_8 only has one element of order 4 and Q_8 has three elements of order 4. Thus they cannot be isomorphic.

Exercise 1.6.7

If $n \neq m$, then we have $n! = |S_n| \neq |S_m| = m!$ so they cannot be isomorphic.

Exercise 1.6.8

Notice that no element of S_4 has order 12 but D_{24} does.

Exercise 1.6.9

Exercise 1.6.10

Exercise 1.6.11

Exercise 1.6.12

Exercise 1.6.13

We use the subgroup criterion. Let $h \in \varphi(G)$. Then there exists $g \in G$ such that $\varphi(g) = h$. Then $h^{-1} = \varphi(g)^{-1} = \varphi(g^{-1}) \in \varphi(G)$. Now suppose that $h, k \in \varphi(G)$. Then there exists $g_1, g_2 \in G$ such that $\varphi(g_1) = h$ and $\varphi(g_2) = k$. Then we have $hk = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) \in \varphi(G)$. Thus $\varphi(G)$ is a subgroup of H.

If φ is injective, then $\varphi : G \to \varphi(G)$ is both injective and surjective and thus is a group isomorphism.

Exercise 1.6.14

Exercise 1.6.15

Exercise 1.6.16

Exercise 1.6.17

Suppose that $\varphi(g) = g^{-1}$ defines a homomorphism. Then we have

$$ghg^{-1}h^{-1} = gh\varphi(g)\varphi(h) = gh\varphi(gh) = gh(gh)^{-1} = 1_G$$

for any $g, h \in G$. Thus G is abelian. Now suppose that G is abelian. Let $g, h \in G$. Then we have

$$\varphi(gh) = (gh)^{-1} = h^{-1}g^{-1} = \varphi(h)\varphi(g) = \varphi(g)\varphi(h)$$

so that φ is a homomorphism.

Exercise 1.6.18

Suppose that $\varphi(g) = g^2$ defines a homomorphism. Let $g, h \in G$. Then we have

$$ghgh = \varphi(gh) = \varphi(g)\varphi(h) = g^2h^2$$

Multiplying by g^{-1} on the left and h^{-1} on the right gives $hg = gh$. Thus G is abelian. Now suppose that G is abelian. Let $g, h \in G$. Then we have

$$\varphi(gh) = ghgh = g^2h^2 = \varphi(g)\varphi(h)$$

Thus φ is a homomorphism.

Exercise 1.6.19

Let $\varphi_k : G \to G$ be the map $\varphi_k(z) = z^k$. I claim that φ_k is a homomorphism. Indeed, we have

$$\varphi_k(zw) = (zw)^k = z^kw^k = \varphi_k(z)\varphi_k(w)$$

for any $z, w \in G$ (multiplication in $\mathbb{C}$ is commutative). Now suppose that $z \in G$ is given by $z = e^{2\pi i/n}$ for some n. Notice that $w = e^{2\pi i/nk} \in G$ because $w^{nk} = 1$. Then clearly we have $\varphi_k(w) = z$. Thus φ is surjective.

When $k > 1$, notice that the kth roots of unity all map to 1 under φ_k. Thus φ_k is not injective.

Exercise 1.6.20

Exercise 1.6.21

Exercise 1.6.22

Let $\varphi_k : A \to A$ be the map $\varphi_k(a) = a^k$. Then we have

$$\varphi_k(ab) = (ab)^k = a^kb^k = \varphi_k(a)\varphi_k(b)$$

for any $a, b \in A$. Hence φ is a homomorphism.

Now suppose that $k = -1$. Notice that φ is surjective since $\varphi(a^{-1}) = a$ for any $a \in A$. Now suppose that $\varphi(a) = \varphi(b)$. Then we have $a^{-1} = b^{-1}$ which implies that $a = b$. Hence φ is also injective.

Exercise 1.6.23

I claim that $G = \{x^{-1}\sigma(x) \mid x \in G\}$. Clearly the set on the right is a subset of G. Moreover, notice that if $x^{-1}\sigma(x) = y^{-1}\sigma(y)$, then we have $\sigma(xy^{-1}) = xy^{-1}$ which implies that $xy^{-1} = 1$. Hence $x = y$. Thus distinct $x \in G$ gives distinct elements $x^{-1}\sigma(x)$. Hence the size of the set on the right is at least $|G|$. Thus we have equality. This means that every element of G is of the form $x^{-1}\sigma(x)$

for some $x \in G$. Now notice that

$$\sigma(x^{-1}\sigma(x)) = \sigma(x)^{-1}\sigma(\sigma(x)) = \sigma(x)^{-1}x$$

Thus σ is the map given by $\sigma(g) = g^{-1}$. Since we assumed that σ is a homomorphism, Q17 implies that G is abelian.

2 Subgroups

2.1

2.2

2.3 Cyclic Groups and Cyclic Subgroups

2.4 Subgroups Generated by Subsets of a Group

Exercise 2.4.1
The smallest subgroup containing H is H itself. Hence $\langle H \rangle = H$.

Exercise 2.4.2

Exercise 2.4.3

Exercise 2.4.4
If $H = \{1_G\}$ then the smallest subgroup of G containing $H \setminus \{1_G\}$ is exactly $\{1_G\} = H$. Hence $\langle H \setminus \{1_G\} \rangle = H$. If $H \neq \{1_G\}$, choose $h \in H$. Then proposition 9 we have $1_G = hh^{-1} \in \langle H \setminus \{1_G\} \rangle$. Thus $H \subseteq \langle H \setminus \{1_G\} \rangle$. Thus H is the smallest subgroup of G containing $H \setminus \{1_G\}$.

Exercise 2.4.5

3 Quotient Groups and Homomorphisms

3.1 Definitions and Examples

Exercise 3.1.1

Exercise 3.1.2

Exercise 3.1.3

Exercise 3.1.4

Exercise 3.1.5

Exercise 3.1.6

Exercise 3.1.7

Exercise 3.1.8

Exercise 3.1.9

Exercise 3.1.10

Exercise 3.1.11

Exercise 3.1.12

Exercise 3.1.13

Exercise 3.1.14

Exercise 3.1.15

Exercise 3.1.16

Let $gN \in G/N$. Since $g \in G = \langle x, y\rangle$, we have $g = x^{n_1}y^{m_1}\cdots x^{n_k}y^{m_k}$ for some $n_1, m_1, \ldots, n_k, m_k \in \mathbb{Z}$. Then we have $gN = x^{n_1}y^{m_1}\cdots x^{n_k}y^{m_k}N = (x^{n_1}N)(y^{m_1}N)\cdots(x^{m_k}N)(y^{m_k}N)$ so that $gN \in \langle xN, yN\rangle$. Hence $G/N = \langle xN, yN\rangle$.

Let $gN \in G/N$. Since $g \in G = \langle S\rangle$, we have $g = s_1^{k_1}\cdots s_m^{k_m}$ for some $s_1, \ldots, s_m \in S$ and $k_1, \ldots, k_m \in \mathbb{Z}$. Then we have $gN = s_1^{k_1}N\cdots s_m^{k_m}N$ so that $G/N = \langle sN \mid s \in S\rangle$.

Exercise 3.1.17

(a) Notice that $s^{i_1}r^{i_2}\langle r^4\rangle = s^{j_1}r^{j_2}\langle r^4\rangle$ if and only if $i_1 = i_2$ and $4 \mid i - j_2$. Thus we get 8 unique elements in $G/\langle r^4\rangle$ given by $1, r, r^2, r^3, s, sr, sr^2, sr^3$.

(b) We have done this above.

(c)

3.2 More on Cosets and Lagrange's Theorem

Exercise 3.2.1

Exercise 3.2.2

Exercise 3.2.3

Exercise 3.2.4
Suppose that G is not abelian and $Z(G) \neq 1$. If $Z(G) = G$ then this is a contradiction. Otherwise, without loss of generality assume that $|Z(G)| = p$. Then $G/Z(G)$ is a group of prime order, and hence is cyclic. I claim that this implies G is abelian. Suppose that $xZ(G)$ generates $G/Z(G)$. Let $g, h \in G$. Then there exists $n, m \in \mathbb{N}$ such that $x^n Z(G) = gZ(G)$ and $x^m Z(G) = hZ(G)$. Then $x^{-n}g, x^{-m}h \in Z(G)$. So there exists $z_1, z_2 \in Z(G)$ such that $g = x^n z_1$ and $h = x^m z_2$. Now we have

$$gh = x^n z_1 x^m z_2 = x^{n+m} z_1 z_2 = x^m z_2 x^n z_1 = hg$$

Thus G is abelian. This is a contradiction.

Exercise 3.2.5

Exercise 3.2.6

Exercise 3.2.7

Exercise 3.2.8

Exercise 3.2.9

Exercise 3.2.10

Exercise 3.2.11

Exercise 3.2.12

Exercise 3.2.13

Exercise 3.2.14

Exercise 3.2.15

Exercise 3.2.16

3.3

3.4 Composition Series and the Holder Program

Exercise 3.4.1
Let $x \in G$. Since G is abelian, we know that $\langle x \rangle$ is a normal subgroup of G. Since G is simple, we must have $G = \langle x \rangle$. Thus G is cyclic. If x does not have finite order, then $\langle x^2 \rangle$ is a strict subgroup of G, a contradiction. Thus G is finite. Thus $G \cong Z_n$ for some $n \in \mathbb{N}^+$. Now if n is composite, say $n = ab$, then $\langle x^a \rangle$ is a strict subgroup of order b, a contradiction. Hence n is prime.

Exercise 3.4.2